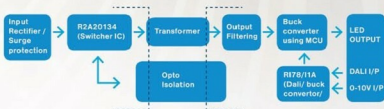


BIG IDEAS
FOR EVERY SPACE

SAVE ENERGY, CREATES COMFORT

RL78/I1A for Fine LED Dimming,
Advanced Safety and Communication



Renesas R2A20134 LED Driver IC and RL78/I1A microcontrollers implement lighting control, advanced safety and communication functions, enabling lower cost, regulation compliance, fine dimming, lighting sequence and color temperature control. RL78/I1A adopts the Renesas 16-bit RL78 CPU, which achieves both high computational performance and ultra-low power consumption.

Key Benefits

- Two Stage solutions offer isolated designs for led driver applications.
- Second stage (Buck convertor) allows flexible output configurations for Vout & Iout
- High resolution dimming and accurate dynamic CRI and CCT control for Lighting Application.
- Supports 16-, 17- & 24-bit extended DALI, DMX512 and multiple sensor & user interface controls.
- Easy integration into building automation network
- Enables high intelligence light innovation at low system cost
- Performance standard as per IEC62384 & safety standard IEC61347-2-13

Applications

- Intelligent LED drivers, Lighting controls

Device Features: RL78 / I1A & R2A20134

- Lighting & Power Control Timer
- PWM High resolution by dithering
- Controlled in-rush current by soft start
- Comparator 6ch
- Window comparator function
- PGA Amp with x4/x8/x16/x32 setting
- Critical Conduction Mode PFC
- Adjustable for Switching Frequency
- Over Current protection
- Low EMI, support Isolated & non-isolated solutions

Specifications

- Input: 100Vac ~ 280Vac
- Output: 30V@1A (Flexible can be tuned to any Output combination
- Current regulation <1%
- Pf > 0.9
- THD <10%
- Target efficiency > 82%
- Dimming Options : 0-10V & Dali
- Surge Capability > 4KV (to be tested)
- Emission: CE/RE as per CISPR 15 (to be tested)
- Dips, Interrupts, Injected current, Flicker (to be tested)